

A

1. (a) The $IQR = 7.4 - 5.8 = 1.6$, so $1.5 \times IQR = 2.4$
A whisker is allowed to extend as much as $1.5 \times IQR$
- below the 1st quartile — or to $5.8 - 2.4 = 3.4$
 - above the 3rd quartile — or to $7.4 + 2.4 = 9.8$
- Since 3.6 is not beyond this (at the low end), it is not considered an outlier.
- (b) When Z-scores are used, we have identified data values with $|Z| \geq 2$ — that is, when $Z \leq -2$ or $Z \geq 2$ — as possible outliers.
- (c) The Z-score for 3.6 is $Z = \frac{3.6 - 6.591}{1.288} \doteq -2.322$.
By this method, then, 3.6 is flagged as an outlier.

2. Only option (a)

3. (a) $P(A^c) = 1 - P(A) = 1 - 0.3 = \underline{0.7}$

(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 0.3 + 0.35 - 0.25 = \underline{0.4}$

(c) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.25}{0.35} \doteq \underline{0.714}$